Jiale Chen

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HomePage | Github | ORCID | Google Scholar | Instagram

Beijing, China

Education

- **Beijing University of Technology** *2023 - 2027*
  B.S. in Electrical and Computer Engineering Beijing, China
  - GPA: 3.75/4.00; Overall Average: 88.44/100; Junior Year Weighted Average: 92.7/100 (Top10)
  - Selected Courses:
    - Advanced Mathematics (100), Control Systems (99), Deep Learning (98), RF Integrated Circuit Design(97), Microcontroller Systems (97), General Physics (97), Digital Intergated Circuit Design (95).
- **Shanghai Jiao Tong University** *2026 - 2027*
  Research Internship Shanghai, China
  - Designed an end-to-end research framework for medical large-model applications, including data processing, model adaptation, multimodal fusion, privacy protection, and evaluation.
  - Investigated blockchain-based privacy protection for medical large-model inference and fine-tuning strategies for multimodal medical models.
  - Conducted few-shot transfer learning on an orbital imaging dataset and reproduced representative state-of-the-art methods for baseline comparison and framework validation.

Publications

[1] **Jiale Chen**.
Communication-Efficient Compression Method for FedDyn Federated Learning Models. []
IEEE 4th International Conference on Control, Electronics and Computer Technology (ICCECT), 2026. Published in IEEE Xplore.

[2] Yihong Chen*, **Jiale Chen***, Hongyu Zhao, Lili Cao, Kaili Qin, Yuan Qiu, Ruhui Ma**, Yuan Liu.
MedChain: Blockchain-based Trusted Evidence Preservation for Medical LLM Inference. []
International Conference on Blockchain and Web3 Technology Innovation and Applications (BWTAC).

Internship

- **Lenovo [][]** *Jul 2025 - Sep 2025*
  Embedded Algorithm Engineer Beijing, China
  - Optimized and deployed Stable Diffusion 3 (SD3) for text-to-image, image-to-image, and inpainting tasks, improving inference latency.
  - Built a segmentation-guided image editing pipeline integrating SD3 and SAM2, enabling automatic image segmentation and high-quality inpainting.
  - Leveraged OpenVINO to deploy and accelerate SD3 on Intel platform, achieving efficient inference performance.
- **Cisco []** *Jan 2026 - Feb 2026*
  Technical Engineer Intern Beijing, China
  - Learns fundamental concepts of AI data center networking, including RDMA over Converged Ethernet (RoCE) and ECN-based congestion control mechanisms.
  - Assists in reviewing switch architectures and network designs for large-scale distributed training scenarios.
  - Supports basic testing and validation tasks for network switches and optical modules under the guidance of engineers.
  - Studies emerging networking technologies and related topics, including programmable networking and reinforcement learning concepts.

Research experience

- **Communication-Efficient Model Compression for FedDyn in Federated Learning** *Dec 2025 - Mar 2026*
  Advisor: Prof.Ruhui Ma | Shanghai Jiao Tong University [ ]
  - Studied communication efficiency challenges in FedDyn under heterogeneous federated learning settings.
  - Analyzed the trade-off between parameter transmission cost and convergence stability.
  - Developed a FedDyn-aware parameter importance evaluation strategy to enable selective communication, effectively reducing bandwidth usage without sacrificing model performance.

- **FPGA-Based Remote Laboratory Platform** *Mar 2025 - Aug 2025*
  Advisor: Prof.Sujuan Liu | Beijing University of Technology [ ]
  - Designed and implemented a cloud-enabled FPGA remote laboratory system based on FPGA, Raspberry Pi 5, and STM32, enabling full remote experiment workflow.
  - Developed a web-based control platform integrating remote JTAG download, logic analyzer, virtual oscilloscope, digital twin, and real-time IP camera monitoring.
  - Implemented simulated JTAG bitstream download via UART (115200 baud), achieving 10–20s remote deployment latency and stable hardware configuration.
  - Built a real-time signal acquisition and debugging framework using FPGA AD/DA modules and protocol-level analysis (UART/I2C/SPI), enabling remote waveform visualization and system-level validation.

- **Efficient and Trustworthy Multimodal Large Language Models** *Mar 2026 - ?*
  Advisor: Hongyu Zhao | Shanghai Jiao Tong University
  - Developed **MedFusion**, a sample-level shared LoRA adaptation framework for multi-source multimodal learning, using fused image-query representations for adapter grouping and dynamic routing.
  - Developed **MedChain**, a blockchain-based trusted evidence preservation framework for medical LLM inference, integrating context-bound evidence modeling with privacy-preserving on-chain/off-chain anchoring and tamper verification.
  - Currently investigating **multimodal LLM quantization and parameter-efficient adaptation**, focusing on low-bit quantization, QLoRA, memory-efficient fine-tuning, and efficient deployment.

Skills

- **Languages:** IELTS: 7.5 (Listening: 8.0, Reading: 8.5). CET-4: 604. CET-6:513.
- **Programming Languages:** Python, C, C++, Verilog, CUDA.
- **AI & Machine Learning:** PyTorch, Hugging Face Transformers, PEFT, OpenCV.
- **Hardware & EDA Tools:** Vivado, Quartus II, ModelSim, Cadence Virtuoso, Keil5, STM32CubeMX.
- **Development Tools:** Linux, Git, Slurm.

Honors and Awards

- **Second Prize, National Undergraduate IC Innovation and Entrepreneurship Competition** *Aug 2025*
  Talent Exchange Center of the Ministry of Industry and Information Technology (MIIT), P.R. China [ ]
- **First Prize, Beijing Integrated Circuit Design Competition (Digital Track)** *May 2026*
  Beijing Municipal Education Commission
- **Third Prize, Chinese Collegiate Computing Competition (National Final)** *Jul 2026*
  Organizing Committee of the Chinese Collegiate Computing Competition (4C), P.R. China [ ]
- **Second Prize, China Undergraduate Mathematical Contest in Modeling (Beijing Region)** *Nov 2025*
  China Society for Industrial and Applied Mathematics (CSIAM) [ ]

MISC

- **Hobbies:** Football, Guitar, and Singing.
- I would like to visit different countries and cities, enjoying multiple culture and exploring novel fields.